

Coriolis Force and Christoffel Symbols

Lagrangian of a Classical Particle

Action,

$$S = \int_{t_1}^{t_2} dt L \quad (1)$$

where

$$L = \frac{m}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu - V(x) \quad (2)$$

By Euler-Lagrange equations of motion:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}^\mu} \right) = \frac{\partial L}{\partial x^\mu} \quad (3)$$

$$\frac{d}{dt} [m g_{\mu\nu} \dot{x}^\nu] = \frac{m}{2} g_{\alpha\nu,\mu} \dot{x}^\alpha \dot{x}^\nu - \partial_\mu V \quad (4)$$

(where $g_{\alpha\beta,\mu} = \partial_\mu g_{\alpha\beta}$)

$$m g_{\mu\nu,\alpha} \dot{x}^\nu \dot{x}^\alpha + m g_{\mu\nu} \ddot{x}^\nu - \frac{m}{2} g_{\alpha\nu,\mu} \dot{x}^\alpha \dot{x}^\nu = -\partial_\mu V \quad (5)$$

Rearranging the terms:

$$g_{\mu\nu} \ddot{x}^\nu + \frac{1}{2} [g_{\mu\nu,\alpha} + g_{\nu\mu,\alpha} - g_{\alpha\nu,\mu}] \dot{x}^\alpha \dot{x}^\nu = -\partial_\mu V \quad (6)$$

Multiplying by $g^{\gamma\mu}$:

$$\ddot{x}^\gamma + g^{\gamma\mu} \frac{1}{2} [g_{\mu\nu,\alpha} + g_{\nu\mu,\alpha} - g_{\alpha\nu,\mu}] \dot{x}^\alpha \dot{x}^\nu = -g^{\gamma\mu} \partial_\mu V \quad (7)$$

These are exactly the Christoffel symbols $\Gamma_{\alpha\nu}^\gamma$.

Equation of Motion

The equation of motion looks like:

$$\ddot{x}^\gamma + \Gamma_{\alpha\nu}^\gamma \dot{x}^\alpha \dot{x}^\nu = -\partial^\gamma V = -g^{\gamma\mu} \nabla_\mu V \quad (8)$$

Notice that the action is *not* invariant under reparametrisation, even affine reparametrisation is not allowed.

Therefore, when the parameter chosen is the ‘time’ experienced by the particle, this equation of motion looks like:

$$\ddot{x}^\gamma + \Gamma_{\alpha\nu}^\gamma \dot{x}^\alpha \dot{x}^\nu = -g^{\gamma\mu} \nabla_\mu V \quad (9)$$

Let $c(t)$ be the curve traversed by the particle. Here the parameter ‘ t ’ is the time experienced by the particle.

Newton’s equation in tensorial form looks like:

$$\nabla_{\frac{\partial}{\partial t}} \frac{\partial}{\partial t} = -g^{\gamma\mu} \partial_\mu V \implies \nabla_{c'(t)} c'(t) = -g^{\gamma\mu} \partial_\mu V \quad (10)$$

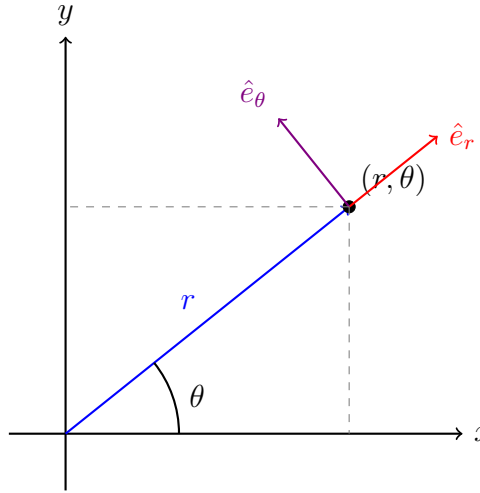
i.e. Covariant derivative of velocity / tangent of the curve, along the curve (i.e. the acceleration) is given by $-g^{\gamma\mu} \partial_\mu V$.

True acceleration is **not** $\frac{d^2 x^\mu}{dt^2}$. Rather it’s $\frac{d^2 x^\mu}{dt^2} + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta$. $\nabla_{c'(t)} c'(t)$ represents the rate of change of tangent vector $c'(t)$ along the curve $c(t)$.

If $\nabla_\mu V = 0 \implies \nabla_t c'(t) = 0$, which is the equation for a geodesic, i.e., without any influence of force, the particle follows a geodesic equation.

Example: Particle on a 2D Plane

Consider the case of a particle undergoing motion on a plane (2D). Instead of Cartesian coordinates, let’s discuss polar coordinates.



In absence of any potential V , $\nabla_t c'(t) = 0 \implies \ddot{x}^\mu + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0$.

For polar coordinates, the only non-zero Christoffel symbols are:

$$\Gamma_{\theta\theta}^r = -r, \quad \Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{r} \quad (11)$$

where $x^\mu(t) = c(t)$ is the coordinate representation of curve c .

The equations look like:

$$\ddot{r} - r\dot{\theta}^2 = 0 \quad (12)$$

$$\ddot{\theta} + \frac{2}{r}\dot{\theta}\dot{r} = 0 \implies r\ddot{\theta} + 2\dot{\theta}\dot{r} = 0 \quad (13)$$

(where $r \neq 0$).

Notice the emergence of centrifugal term $r\dot{\theta}^2$ and Coriolis term $2\dot{\theta}\dot{r}$ because of Christoffel symbols for this given coordinate system.

Now, let’s introduce a potential $V(r, \theta)$ in the picture:

$$\ddot{r} - r\dot{\theta}^2 = -g^{rr} \partial_r V - g^{r\theta} \partial_\theta V \quad (14)$$

$$\ddot{\theta} + \frac{2}{r}\dot{\theta}\dot{r} = -g^{\theta r}\partial_r V - g^{\theta\theta}\partial_\theta V \quad (15)$$

Using the metric:

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \quad g^{\alpha\beta} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{r^2} \end{pmatrix} \quad (16)$$

This gives:

$$\ddot{r} - r\dot{\theta}^2 = F_r \quad [F_r = -\partial_r V, F_\theta = -\partial_\theta V] \quad (17)$$

$$\ddot{\theta} + \frac{2}{r}\dot{\theta}\dot{r} = \frac{1}{r^2}F_\theta \quad (18)$$

Rewriting:

$$m\ddot{r} = F_r + mr\dot{\theta}^2 \quad (\text{Centrifugal force}) \quad (19)$$

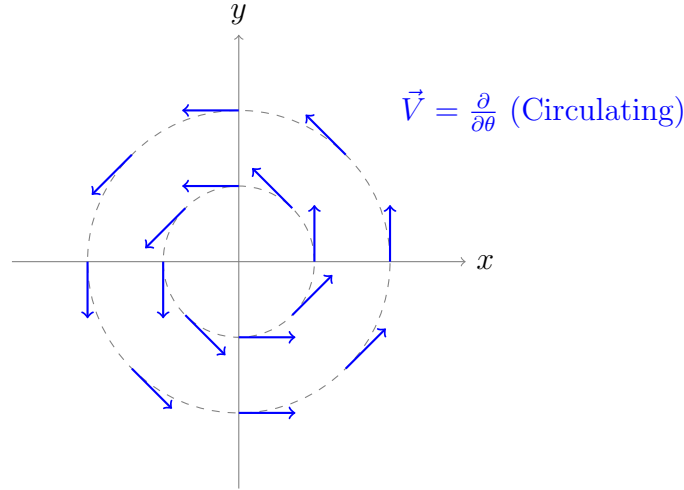
$$mr^2\ddot{\theta} + 2mr\dot{r}\dot{\theta} = F_\theta \implies mr\ddot{\theta} = \frac{F_\theta}{r} - 2m\dot{r}\dot{\theta} \quad (\text{Coriolis force}) \quad (20)$$

We now see this fictitious force show up because of Christoffel symbols, i.e., because of coordinate transformation.

Christoffel Symbol as Coefficients of Basis Rotation

Consider a vector field $V = \sin(\theta)\hat{e}_r$, described in polar coordinates as $V^r = 0$, $V^\theta = 1$. Naively one might say that this represents a ‘constant’ vector field on $\mathbb{R}^2$, i.e. acceleration is zero, i.e. along any curve $\frac{d}{dt}V^\mu = 0$.

But it seems obviously wrong, because it represents a velocity field circulating around the origin, whose acceleration in Cartesian coordinates is non-zero.



Therefore, one also needs a coordinate-independent description of acceleration, because $\frac{dV^\mu}{dt}$ only accounts for change in component of V along x^μ as one moves along $c(t)$.

To begin with, let's look at the same vector field in Euclidean coordinates:

$$V = V^r \frac{\partial}{\partial r} + V^\theta \frac{\partial}{\partial \theta} = \frac{\partial}{\partial \theta} \quad (21)$$

In this coordinate, acceleration is non-zero because of the rotation of this basis vector e_θ , and is given by $-r$ in radial direction.

We see that in curved coordinate systems, one needs to also take into account the change of basis vectors (rotation of basis vectors) as one moves along the curve.

i.e. Total rate of change:

$$\frac{DV^\alpha}{Dt} = \frac{dV^\alpha}{dt} + \left(V^\mu \frac{de_\mu}{dt} \right)^\alpha \quad (22)$$

Represent:

$$\frac{de_\mu}{dt} = \Gamma_{\mu\nu}^\alpha \dot{x}^\nu e_\alpha \quad (23)$$

Call $\frac{D}{Dt}$ along curve c as total derivative of vector field V along c .

$$\nabla_{\frac{\partial}{\partial t}} V \equiv \nabla_{c'(t)} V \quad (24)$$

$$\left(\nabla_{\frac{\partial}{\partial t}} V \right)^\mu = \frac{dV^\mu}{dt} + \Gamma_{\alpha\beta}^\mu V^\alpha \dot{x}^\beta \quad (25)$$

$\implies$ Christoffel symbols take care of the rate of change of basis vectors while traversing along a curve $c(t)$ in a curved manifold.